

Estimation (Part 3)

STAT 4530/6530
Statistical Inference for Data Scientists

Constructing Confidence Intervals

In a study that employs randomization, let $T_L(\mathbf{Y})$ and $T_U(\mathbf{Y})$ be two functions of observations $\mathbf{Y} = (Y_1, \dots, Y_n)$ such that

$$P[T_L(\mathbf{Y}) \leq \theta \leq T_U(\mathbf{Y})] = 1 - \alpha$$

for all possible values of θ . Once the data $\mathbf{Y} = \mathbf{y}$ are observed, the interval $[T_L(\mathbf{y}), T_U(\mathbf{y})]$ is called a $100(1 - \alpha)\%$ **confidence interval** for θ with lower and upper **confidence limits** $T_L(\mathbf{y})$ and $T_U(\mathbf{y})$, respectively.

The probability $1 - \alpha$ is called the **confidence level** and is usually chosen to be close to 1 (such as 0.95).

The corresponding probability α is called the **error probability**.

Constructing Confidence Intervals

Note the distinction between “confidence” and “probability”

- We assume that the parameter θ has some unknown but fixed value
- Once we have observed the data, the confidence interval either does or does not contain θ . It is impossible for there to be, for example, a 0.95 probability that the interval contains θ .
- A probability of 0.95 can apply to the random interval $[T_L(\mathbf{Y}), T_U(\mathbf{Y})]$, with endpoints that are random variables, **before** we observe the data.
- The interpretation of a 95% confidence interval is that if we repeatedly observe new data samples and construct a 95% confidence interval from each sample of observed data, over the long run about 95% of such intervals will contain θ .

Constructing Confidence Intervals

We can often find confidence limits $T_L(\mathbf{y})$ and $T_U(\mathbf{y})$ using the sampling distribution of a statistic $T(\mathbf{Y})$ that is a good estimator of θ .

For example, if $T(\mathbf{Y})$ has a $N(\theta, \sigma_T^2)$ distribution, then

$$\frac{T(\mathbf{Y}) - \theta}{\sigma_T} \sim N(0, 1).$$

The ratio above, which is a function of $\mathbf{Y}$ and θ but has a distribution that does not depend on θ , is called the **pivotal quantity**.

Constructing Confidence Intervals

Because 95% of a $N(0, 1)$ distribution's density is between -1.96 and 1.96,

$$P \left[-1.96 \leq \frac{T(\mathbf{Y}) - \theta}{\sigma_T} \leq 1.96 \right] = 0.95,$$

which implies that

$$P [T(\mathbf{Y}) - 1.96\sigma_T \leq \theta \leq T(\mathbf{Y}) + 1.96\sigma_T] = 0.95.$$

Once we actually observe the data $\mathbf{y}$, the interval $(T(\mathbf{y}) - 1.96\sigma_T, T(\mathbf{y}) + 1.96\sigma_T)$ is a 95% confidence interval for θ . We also express this interval as $T(\mathbf{y}) \pm 1.96\sigma_T$.

More generally, for confidence level $1 - \alpha$, the confidence interval uses standard normal quantiles with right-tail probabilities equal to $\alpha/2$, denoted by $z_{\alpha/2}$, and has the form $(T(\mathbf{y}) - z_{\alpha/2}\sigma_T, T(\mathbf{y}) + z_{\alpha/2}\sigma_T)$. For example, $z_{0.05/2} = z_{0.025} = 1.96$ and $z_{0.01/2} = z_{0.005} = 2.58$.

A Large-Sample Confidence Interval for the Mean

Consider constructing a confidence interval for a population mean of a quantitative variable using n independent observations from a population with mean μ and variance σ^2 , such as in a randomized study.

By the central limit theorem, for large n the sampling distribution of the sample mean $\bar{Y}$ is approximately normal with mean μ and standard error $\sigma/\sqrt{n}$. Therefore

$$Z = \frac{\bar{Y} - \mu}{\sigma/\sqrt{n}}$$

has an approximate standard normal distribution. Using the observed sample mean $\bar{y}$, $\bar{y} \pm 1.96\sigma/\sqrt{n}$ is a 95% confidence interval for μ .

A Large-Sample Confidence Interval for the Mean

By the central limit theorem, for large n the sampling distribution of the sample mean $\bar{Y}$ is approximately normal with mean μ and standard error $\sigma/\sqrt{n}$. Therefore

$$Z = \frac{\bar{Y} - \mu}{\sigma/\sqrt{n}}$$

has an approximate standard normal distribution. Using the observed sample mean $\bar{y}$, $\bar{y} \pm 1.96\sigma/\sqrt{n}$ is a 95% confidence interval for μ .

Problem: We do not know σ .

Solution: The method is designed for large n , in which case the sample standard deviation

$$s = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (y_i - \bar{y})^2}$$

is close to σ , we could approximate the confidence interval as $\bar{y} \pm 1.96(s/\sqrt{n})$.

A Large-Sample Confidence Interval for the Mean

By the central limit theorem, for large n the sampling distribution of the sample mean $\bar{Y}$ is approximately normal with mean μ and standard error $\sigma/\sqrt{n}$. Therefore

$$Z = \frac{\bar{Y} - \mu}{\sigma/\sqrt{n}}$$

has an approximate standard normal distribution. Using the observed sample mean $\bar{y}$, $\bar{y} \pm 1.96\sigma/\sqrt{n}$ is a 95% confidence interval for μ .

Problem: We do not know σ .

Solution: The method is designed for large n , in which case the sample standard deviation

$$s = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (y_i - \bar{y})^2}$$

is close to σ , we could approximate the confidence interval as $\bar{y} \pm 1.96(s/\sqrt{n})$.

Better Solution: Later, we will discuss a better approach that accounts for the error introduced by substituting s for σ .

Confidence Intervals for Proportions

Consider independent binary observations arising from a $\text{Ber}(p)$ distribution. For large n , the sampling distribution of $\bar{Y}$ is approximately normal with mean p and standard error $\sqrt{p(1-p)/n}$. The pivotal quantity is

$$Z = \frac{\bar{Y} - p}{\sqrt{\frac{p(1-p)}{n}}},$$

which has an approximate standard normal distribution.

To construct a confidence interval for p after observing $\bar{y}$, we solve the equalities

$$\bar{y} - p \pm z_{\alpha/2} \sqrt{\frac{p(1-p)}{n}}$$

for p to get the endpoints.

Confidence Intervals for Proportions

To construct a confidence interval for p after observing $\bar{y}$, we solve the equalities

$$\bar{y} - p \pm z_{\alpha/2} \sqrt{\frac{p(1-p)}{n}}$$

for p to get the endpoints.

We can square both sides and solve the resulting equation, which is quadratic in p , to obtain the confidence interval

$$\left(\bar{y} + \frac{z_{\alpha/2}^2}{2n} \pm z_{\alpha/2} \sqrt{(\bar{y}(1-\bar{y}) + z_{\alpha/2}^2/4n) / n} \right) / \left(1 + \frac{z_{\alpha/2}^2}{n} \right),$$

called the **score confidence interval** for a proportion.

Although the theory is based on a large sample size n , it performs reasonably well for any n encountered in practice.

Confidence Intervals for Proportions

$$\left(\bar{y} + \frac{z_{\alpha/2}^2}{2n} \pm z_{\alpha/2} \sqrt{\left(\bar{y}(1 - \bar{y}) + \frac{z_{\alpha/2}^2}{4n} \right) / n} \right) / \left(1 + \frac{z_{\alpha/2}^2}{n} \right)$$

The formula for the score confidence interval above is somewhat messy. To simplify, note that as n gets larger, $\frac{z_{\alpha/2}^2}{n}$ approaches 0 and $1 + \frac{z_{\alpha/2}^2}{n}$ approaches 1. This yields the approximate formula

$$\bar{y} \pm z_{\alpha/2} \sqrt{\frac{\bar{y}(1 - \bar{y})}{n}}.$$

This is known as the **Wald confidence interval**. While the simplicity of the formula is appealing, the Wald confidence interval does not perform as well as the score confidence interval. For example, the actual probability that a 95% Wald confidence interval contains the true parameter value p can be much less than 0.95, especially when p is close to 0 or 1.

Determining the Sample Size before Collecting Data

The margin of error for a confidence interval depends on the standard error of the point estimate.

- Thus, the basis for inferential power lies in the formulas for the standard errors
- Before the data collection begins, a study should determine the sample size that will provide the desired degree of precision in estimating the parameter of main interest.
- A relevant measure is the value n for which the confidence interval has margin of error equal to some specified value.
- The key results for finding the sample size are that the margin of error depends directly on the standard error of the sampling distribution of the point estimator, and the standard error itself depends on the sample size.

Determining the Sample Size before Collecting Data

Example: An entrepreneur plans to open a new restaurant and wants to mail an advertisement to a random sample of homes in their city. The ad contains a coupon that would need to be redeemed within two weeks.

- To cut costs, they do not want to use a very large mailing list until they are able to determine whether this method of advertising is effective, with at least 10% of the households that receive the coupon actually using it. How large of a mailing list would be necessary to make this determination?
- They decide to use a sample size n for the mailing list such that, with probability 0.95, the error in estimating the proportion p of households in the city that would redeem such a coupon would not exceed 0.04.
- Thus, if we model the outcome Y_i for each household on the list as binary variable with a $\text{Ber}(p)$ distribution (with redemption of the coupon coded as a 1), we want n such that a 95% confidence interval for p equals $\bar{Y} \pm 0.04$.

Determining the Sample Size before Collecting Data

Because the sampling distribution of $\bar{Y}$ is approximately normal, $\bar{y}$ falls within 1.96 standard errors of p with probability 0.95.

- Thus if n is such that 1.96 standard errors equals 0.04, then with probability 0.95, $\bar{Y}$ falls within 0.04 units of p .
- The standard error is $\sqrt{p(1-p)/n}$, so we set

$$0.04 = 1.96 \sqrt{p(1-p)/n}$$

and solve for n :

$$n = \frac{(1.96)^2 p(1-p)}{(0.04)^2}.$$

Determining the Sample Size before Collecting Data

- The standard error is $\sqrt{p(1-p)/n}$, so we set

$$0.04 = 1.96\sqrt{p(1-p)/n}$$

and solve for n :

$$n = \frac{(1.96)^2 p(1-p)}{(0.04)^2}.$$

- We do not know p , but largest possible value for $p(1-p)$ is achieved when $p = 0.50$. Thus, plugging $p = 0.50$ in formula above to get $n = 600.25$ (and then using a sample size of 601) would ensure that the margin of error will not exceed 0.04, no matter what the true value of p is.
- This will be overly conservative (and n excessively large) if p is not near 0.50. An alternative approach would be to use an estimate of p based upon external information (such as experiences of other who have tried similar mailings).

Confidence Intervals for Means of Normal Populations

For constructing confidence intervals for means,

$$Z = \frac{\bar{Y} - \mu}{\sigma / \sqrt{n}}$$

is a pivotal quantity for large samples, in which case it has approximately a standard normal distribution (meaning a $N(0, 1)$ distribution).

If our samples are actually taken from a normal population distribution, then Z has an exact standard normal distribution, no matter what the size of the sample is.

In any case, σ is almost always an unknown parameter, so we substitute the sample standard deviation

$$s = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (y_i - \bar{y})^2}$$

for σ . However, this can introduce substantial additional error when the sample size n is small.

Confidence Intervals for Means of Normal Populations

Let S denote the random variable version of s :

$$S = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (Y_i - \bar{Y})^2}.$$

For independent observations from a normal distribution, the pivotal quantity

$$T = \frac{\bar{Y} - \mu}{S/\sqrt{n}}$$

has a sampling distribution that is slightly more spread out than the standard normal distribution. This distribution is known as the **t distribution**. It also called **Student's t distribution**.

Confidence Intervals for Means of Normal Populations

The t distribution has the following notable properties:

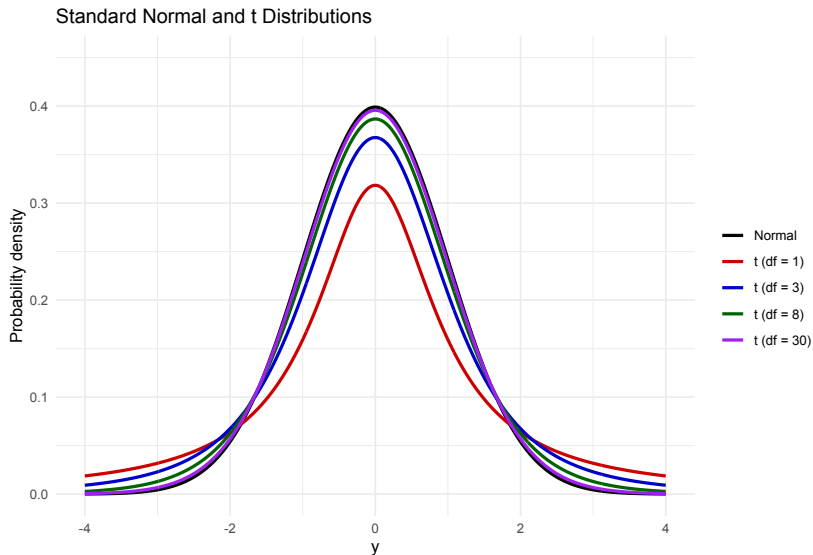
- The t distribution has support on the real line, is bell-shaped, and is symmetric about 0
- The pdf of the t distribution is proportional to $1/[1 + (y^2/d)]^{(d+1)/2}$ (meaning that it is equal to $1/[1 + (y^2/d)]^{(d+1)/2}$ times a normalizing constant that ensures that integrates to 1).
- The index d in the pdf is called the **degrees of freedom** and is denoted by " df ." For inference about a single population mean, $df = n - 1$.
- The t distribution has mean of 0 when $df > 1$ (otherwise, its mean is undefined)
- When $df > 2$, the variance is $df/(df - 2)$, when $1 < df \leq 2$, the variance is infinite, and when $df \leq 1$, the variance is undefined

Confidence Intervals for Means of Normal Populations

The t distribution has the following notable properties (continued):

- The t distribution has thicker tails and is more spread out than the standard normal distribution
- The larger the df value, the more closely the t distribution resembles the standard normal distribution
- When df is about 30 or higher, the two distributions are practically identical

Confidence Intervals for Means of Normal Populations



Confidence Interval for a Mean Using the t Distribution

Let $t_{a,df}$ denote the quantile for a t distribution with df degrees of freedom having probability a in the right-tail (in other words, the $1 - a$ quantile).

For n independent observations from a normal population distribution with sample mean $\bar{y}$, a $100(1 - \alpha)\%$ confidence interval for the population mean μ is

$$\bar{y} \pm t_{\alpha/2, n-1} * \frac{s}{\sqrt{n}}.$$

Example: Estimating Weight Change for Anorexic Girls

An experimental study compared three therapies for young girls suffering from anorexia, an eating disorder.

For each girl, weight was measured before and after a fixed period of treatment designed to aid weight gain. For 29 girls who underwent the cognitive behavioral therapy, the weights at the end of the study minus the weights at the beginning of the study (in pounds) were:

1.7, 0.7, -0.1, -0.7, -3.5, 14.9, 3.5, 17.1, -7.6, 1.6, 11.7, 6.1, 1.1, -4.0, 20.9, -9.3, 2.1,
1.4, -0.3, -3.7, -1.4, -0.8, 2.4, 12.6, 1.9, 3.9, 0.1, 15.4, -0.7

Example: Estimating Weight Change for Anorexic Girls

For 29 girls who underwent the cognitive behavioral therapy, the weights at the end of the study minus the weights at the beginning of the study (in pounds) were:

1.7, 0.7, -0.1, -0.7, -3.5, 14.9, 3.5, 17.1, -7.6, 1.6, 11.7, 6.1, 1.1, -4.0, 20.9, -9.3, 2.1,
1.4, -0.3, -3.7, -1.4, -0.8, 2.4, 12.6, 1.9, 3.9, 0.1, 15.4, -0.7

The weight changes had $\bar{y} = 3.007$ pounds, with $s = 7.309$, so the estimated standard error is $s/\sqrt{n} = 7.309/\sqrt{29} = 1.357$.

A 95% confidence interval for the mean μ uses $t_{0.025,28} = 2.048$, leading to the interval

$$\bar{y} \pm t_{0.025,28} * \frac{s}{\sqrt{n}} = 3.007 \pm 2.048 * 1.357 = (0.23, 5.79).$$

Example: Estimating Weight Change for Anorexic Girls

A 95% confidence interval for the mean μ is (0.23, 5.79). Based on this, it appears that μ is positive, but it may be very small.

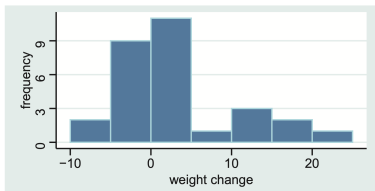
To be safer, we could instead compute a 99% confidence interval. It uses $t_{0.005,28} = 2.763$, yielding

$$\bar{y} \pm t_{0.025,28} * \frac{s}{\sqrt{n}} = 3.007 \pm 2.763 * 1.357 = (-0.74, 6.76).$$

This confidence interval is wider, the cost of having greater confidence. It contains 0, suggesting that 0 is a plausible value for μ .

Another reason for caution is that this experimental study used a volunteer sample, because it is not possible to identify and randomly sample a population of anorexic girls. Inferences are tentative, and “95% confidence” or “99% confidence” in results may be overly optimistic.

Example: Estimating Weight Change for Anorexic Girls



Another caveat is shown in the above histogram of the sample: the sample data distribution of weight changes is skewed to the right. Thus the assumption of a normal population distribution may be violated (more on that shortly).

The median weight change is only 1.40 pounds, less than the mean of 3.01 pounds because of the rightward skew. The sample median provides more evidence that the size of any true treatment effect could be quite small.

Robustness

The t confidence interval for a mean assumes independent observations from a normal population distribution.

For the anorexia study, the histogram is not a precise depiction of the population distribution because the sample size is just 29. Nevertheless it shows evidence of skewness. In general, the normal population distribution assumption seems worrisome, since variables often have population distributions that are quite far from normal.

A statistical method is called **robust** with respect to an assumption if it performs adequately even when that assumption is violated.

The confidence interval for a mean is quite robust against violations of the normal population assumption – even if the population distribution is not normal, t distribution confidence intervals typically still perform quite well, especially when the sample size n exceeds about 15.

Robustness

The confidence interval for a mean is quite robust against violations of the normal population assumption – even if the population distribution is not normal, t distribution confidence intervals typically still perform quite well, especially when the sample size n exceeds about 15.

Moreover, as the sample size n increases, the normal population assumption becomes less important because of the central limit theorem:

- The sampling distribution of $\bar{Y}$ is then bell-shaped even when the population distribution is not.
- The actual probability that the 95% confidence interval method generates an interval that contains μ is close to 0.95 and gets even closer as n increases.

Robustness

An important case when the method does not perform well is when the data are extremely skewed or contain extreme outliers.

Also, the t distribution confidence interval method is not robust to violations of the data-collection assumption. Like all inferential statistical methods, the method has questionable validity if the method for producing the data did not use randomization.

Reference

Agresti, A., & Kateri, M. (2021). Foundations of Statistics for Data Scientists: With R and Python (1st ed.). Chapman and Hall/CRC.